# STRUCTURAL CALCULATION NOTE

East Steel Frame ST-E2 with East Arch Hangers - stand-alone shopfit steel

## تنفيذ متجر مدينة المعرفة 2 - ستايل مديني

Madina 2 Store (Madinah style) - Knowledge Economic City Mall, Madinah

Project No. SIC-043-GEN-GSH2

CLIENT

شركة هدايا الطيبة

Attayyibah Gifts

| Item | Value |
|---|---|
| Client | Attayyibah Gifts |
| Document No. | MD2-SF-ZZ-GF-CA-S-0001 |
| Revision | 00 - Preliminary - for review |
| Date | 12.11.2026 |
| Design codes | SBC 301-2018 (loads), SBC 306-2018 / AISC 360-16 LRFD (steel) |
| Software | Autodesk Robot Structural Analysis Professional 2026 (model Madina2_ST-E2_asbuilt.rtd) |
| Prepared by | Samaya Factory - Technical Office |
| Checked by | Mohamed Sultan |
| Approved by | Client engineer |

# DOCUMENT CONTROL

*Revision history*

| Rev | Date | Status | Change |
|---|---|---|---|
| 00 | 12.11.2026 | Preliminary - for review | First issue: as-built frame of 12 Nov 2026 (lean frame, hung shelf plates, angle hangers with M10 height adjustment) |

*Sign-off*

| Role | Name | Signature | Date |
|---|---|---|---|
| Prepared | Samaya Factory - Technical Office | | 12.11.2026 |
| Checked | Mohamed Sultan | | |
| Approved | Client engineer | | |

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# EXECUTIVE SUMMARY

This note checks the stand-alone steel frame behind the east arches (East Steel Frame ST-E2) and the arch hangers that carry the east shelf, as modelled on 12 Nov 2026, to SBC 301-2018 and SBC 306-2018 (AISC 360-16 LRFD). The frame stands on the floor with no anchors and no wall fixing.

*Table 1 - Key results*

| Item | Value | Limit | Result |
|---|---|---|---|
| Highest member ratio (hanger angles L 40x40x4) | 0.45 | 1.00 | OK |
| Shelf plate deflection, SLS | 2.8 mm | 608/125 = 4.9 mm | OK |
| Front truss deflection, SLS | 2.8 mm | 4209/360 = 11.7 mm | OK |
| Top sway, SLS | 1.0 mm | H/400 = 7.3 mm | OK |
| Stability 0.9D + 1.6N, smallest foot load | 0.18 kN | > 0 (no uplift) | OK |
| Total load on floor, D / L | 3.50 / 5.29 kN | - | - |

## Codes, loads and combinations at a glance

*Table 2 - Design basis summary*

| Item | Value | Reference |
|---|---|---|
| Load code | SBC 301-2018 | - |
| Steel code | SBC 306-2018 / AISC 360-16 LRFD | - |
| Dead load D | self-weight + cladding/pegboards 0.40 kN/m + shelf board 0.25 kN/m | Table 5 |
| Live load L | shelf goods 1.5 kN/m2; pegboard goods 0.5 kN/m2 | SBC 301 Ch. 4 |
| Notional load N | 0.002 x factored gravity, horizontal | SBC 306 C2.2b |
| Strength | 1.2D + 1.6L + 1.6N; 1.4D | SBC 301 2.3.2 |
| Stability | 0.9D + 1.6N | SBC 301 2.3.2 |
| Serviceability | D + L + N | SBC 301 App. C |

1. All members pass SBC 306; the highest ratio is 0.45.
2. All deflections are within the SBC 301 Appendix C limits.
3. Every foot stays in compression in the stability combination, so the frame stands without anchors.
4. The shelf carries 1.5 kN/m2 of goods (about 350 kg over 3.84 m) and the pegboards 0.5 kN/m2.
5. The 40 x 6 hanger flat bars are screwed to both arch heads at 150 mm centres along the seam; this is their buckling restraint.

# DESIGN BASIS

## Codes

*Table 3 - Codes*

| Code | Use |
|---|---|
| SBC 301-2018 | Loads: live loads (Ch. 4), load combinations (2.3.2 LRFD, 2.4 SLS), deflection limits (Appendix C) |
| SBC 306-2018 / AISC 360-16 LRFD | Steel members: tension D2, compression E3, flexure F, combined forces H1-1, notional loads C2.2b, bolts J3.6 |

## Materials

*Table 4 - Materials*

| Item | Grade | Fy (MPa) | E (MPa) |
|---|---|---|---|
| SHS 40x40x2 (cold-formed) | S235 | 235 | 200000 |
| L 40x40x4, flat bars, plates | S235 | 235 | 200000 |
| M10 threaded rods | Grade 4.6 | 240 | 200000 |

## Loads

*Table 5 - Load cases*

| Case | Load | Value | Applied to | Source |
|---|---|---|---|---|
| 1 D | Self-weight of steel | 78.5 kN/m3 | all members | Robot |
| 1 D | Back cladding BK-E + pegboards | 0.40 kN/m | back rail +1.500 | 18 FR MDF + 26 pegboard |
| 1 D | Shelf ES-1 board (36 mm) | 0.25 kN/m | each shelf plate | board weight |
| 2 L | Shelf goods 1.5 kN/m2 x 0.593 m x 1.316 m / 0.608 m | 1.93 kN/m | each shelf plate | SBC 301 Ch. 4 |
| 2 L | Pegboard goods 0.5 kN/m2 x 1.4 m | 0.70 kN/m | back rail +1.500 | SBC 301 Ch. 4 |
| 3 N | Notional 0.002 x factored gravity | 0.007 kN/m | back rail +2.900, horizontal | SBC 306 C2.2b |

## Load combinations

*Table 6 - Load combinations (SBC 301 2.3.2 / 2.4)*

| No. | Combination | Type | Use |
|---|---|---|---|
| 10 | 1.2 D + 1.6 L + 1.6 N | ULS | member strength |
| 12 | 1.4 D | ULS | member strength (dead only) |
| 13 | 0.9 D + 1.6 N | ULS | stability, no uplift at any foot |
| 11 | 1.0 D + 1.0 L + 1.0 N | SLS | deflection |

## Structural system

The frame is 4209 mm long, 588 mm deep and 2920 mm high, in SHS 40x40x2. A front truss (chords at +2.900 and +2.800, five diagonals) spans between two front posts. Four back posts carry back rails at +0.028, +1.500 and +2.900, with one back diagonal and two end frames (top rail, rail at +1.500, floor rail and diagonal). At each of the two arch joints an angle L 40x40x4 spans from the front chord to the back rail; two M10 rods with nuts hang from it (height adjustment) onto 40 x 6 flat bars that run down the seam between the arch heads to the 116 x 10 shelf plate. The

shelf plate is welded to the back joint post and hung at two points. The feet are pinned and rest on the floor without anchors.

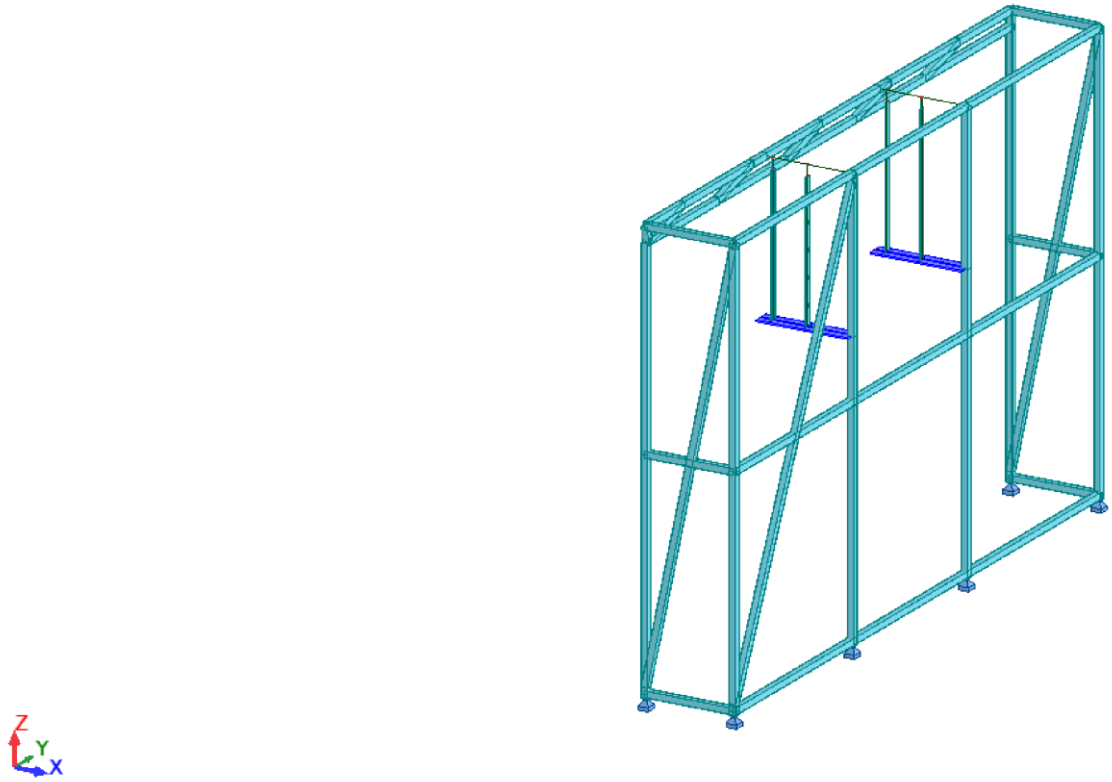



*Figure 1 - Robot analytical model (sections and supports)*

## ANALYSIS

Linear static analysis in Robot Structural Analysis 2026: 56 nodes, 78 bars, 6 pinned feet. Equilibrium: dead load reactions 3.50 kN and live load reactions 5.29 kN, equal to the applied loads.

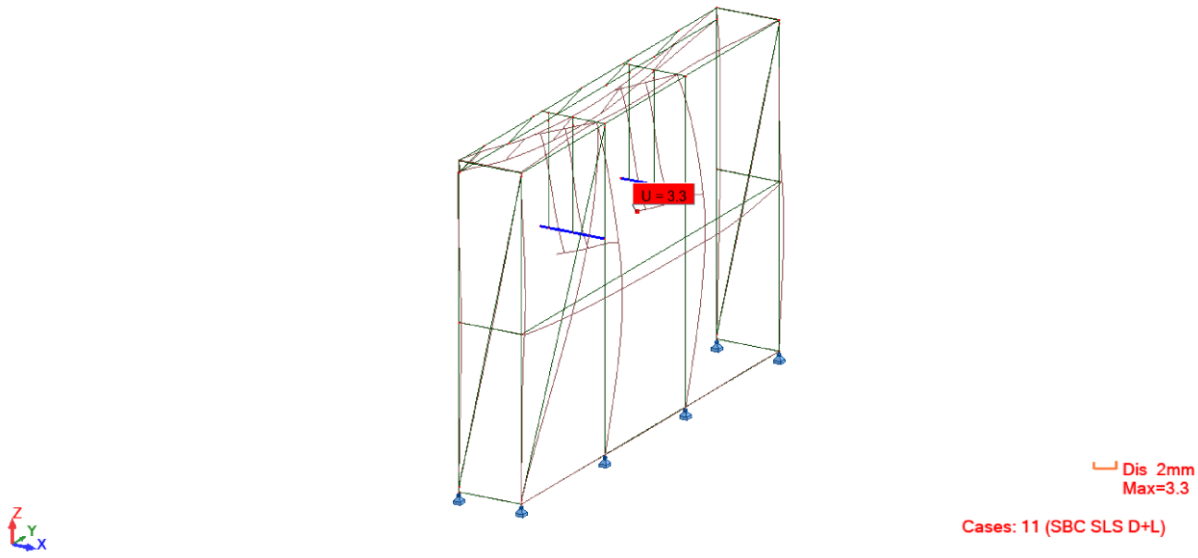



*Figure 2 - SLS deformation, combination 11 (Robot)*

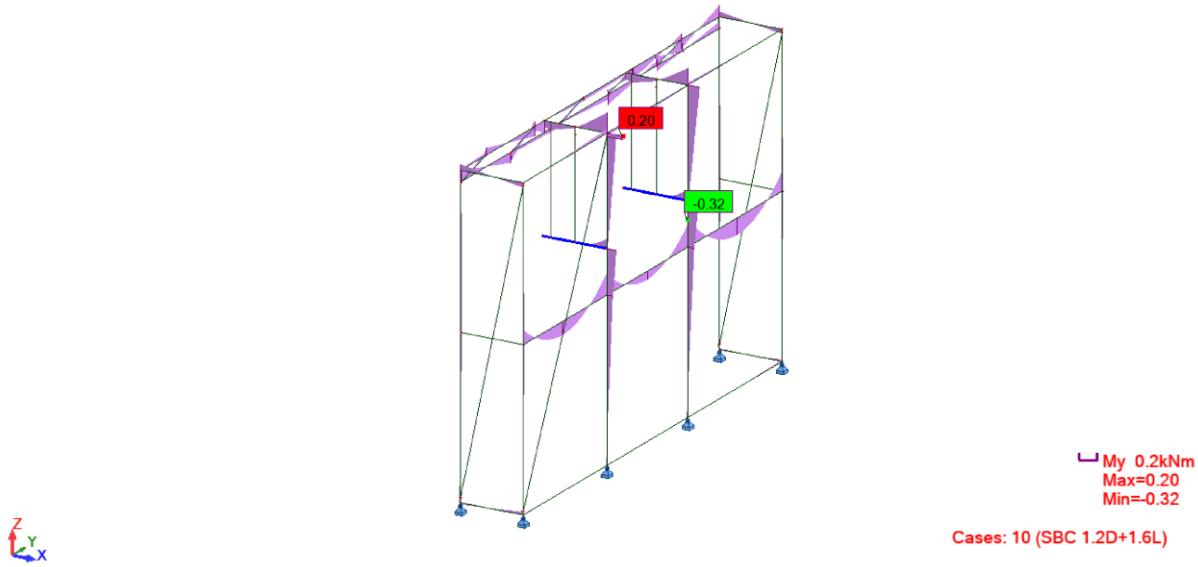



*Figure 3 - Bending moment MY, combination 10 (Robot)*

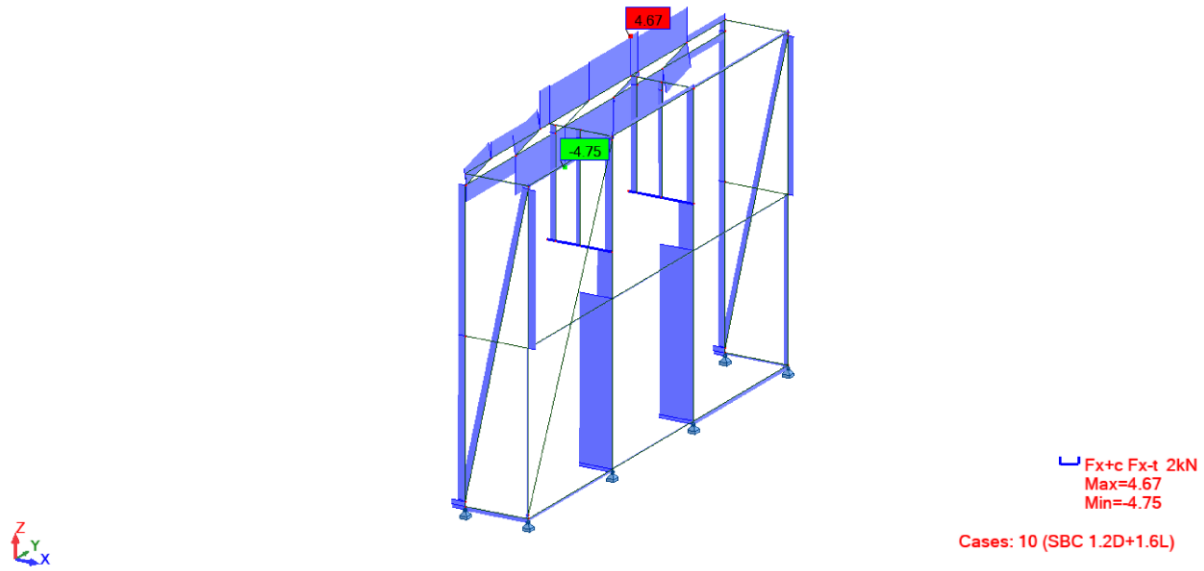



*Figure 4 - Axial force FX, combination 10 (Robot)*

## Support reactions

*Table 7 - Vertical reactions (positive = foot pressing on the floor)*

| Foot | D (kN) | L (kN) | ULS 10 (kN) | Stab. 13 (kN) | Result |
|---|---|---|---|---|---|
| x 5750 / y 776 | 0.41 | 0.66 | 1.66 | 0.49 | OK |
| x 5750 / y 4985 | 0.38 | 0.56 | 1.46 | 0.45 | OK |
| x 6338 / y 776 | 0.33 | 0.06 | 0.38 | 0.18 | OK |
| x 6338 / y 2138 | 1.00 | 1.88 | 4.20 | 0.90 | OK |
| x 6338 / y 3454 | 1.02 | 1.95 | 4.33 | 0.91 | OK |
| x 6338 / y 4985 | 0.36 | 0.20 | 0.63 | 0.21 | OK |

## MEMBER CHECKS

SBC 306 (AISC 360-16 LRFD), phi = 0.90 for tension, compression and flexure. Compression strength Fcr from E3 with K = 1.0 over the braced length (back rails between posts; flat bars between screws at 150 mm); flexure with the plastic modulus (compact sections). Interaction H1-1a when Pr/Pc >= 0.2, else H1-1b. Forces are the governing envelope of combinations 10 and 12.

*Table 8 - Member checks*

| Group | Section | No. | Governing | N (kN) | M (kNm) | KL/r | Ratio | Result | Clause |
|---|---|---|---|---|---|---|---|---|---|
| Hanger angles | L 40x40x4 | 2 | HA1 | 0.08 | 0.24 | 25 | 0.45 | OK | F, H1-1b |
| Back rails | SHS 40x40x2 | 3 | BR1500 | -0.15 | 0.24 | 86 | 0.33 | OK | E3, H1-1b |
| Back posts | SHS 40x40x2 | 4 | BP2 | 0.92 | 0.20 | 61 | 0.28 | OK | D2, H1-1b |
| Front truss top chord | SHS 40x40x2 | 1 | TC | 4.53 | 0.13 | 50 | 0.25 | OK | D2, H1-1a |
| Front truss diagonals | SHS 40x40x2 | 5 | TD9 | -3.23 | 0.17 | 26 | 0.23 | OK | E3, H1-1a |
| Shelf plates (hung) | PL 116x10 | 2 | SP2 | 0.01 | 0.13 | - | 0.21 | OK | F11 |
| Front truss bottom chord | SHS 40x40x2 | 1 | BC | -4.14 | 0.12 | 50 | 0.18 | OK | E3, H1-1a |
| End top rails | SHS 40x40x2 | 2 | ER1 | -0.10 | 0.07 | 37 | 0.15 | OK | E3, H1-1b |
| M10 adjusting rods | M10 (As 58 mm2) | 4 | HR1a | -0.75 | 0.00 | 33 | 0.14 | OK | J3.6, H1-1b |
| Front posts | SHS 40x40x2 | 2 | FP2 | 0.61 | 0.10 | 82 | 0.14 | OK | D2, H1-1b |
| Back diagonal | SHS 40x40x2 | 1 | BD1 | 0.07 | 0.09 | 198 | 0.12 | OK | D2, H1-1b |
| End rails +1.500 | SHS 40x40x2 | 2 | EM1 | -0.03 | 0.06 | 37 | 0.07 | OK | E3, H1-1b |
| End floor rails | SHS 40x40x2 | 2 | EF1 | -0.50 | 0.04 | 37 | 0.06 | OK | E3, H1-1b |
| Hanger flat bars (seam) | FB 40x6 | 4 | HF2a | -0.77 | 0.00 | 87 | 0.05 | OK | E3, H1-1b |
| End diagonals | SHS 40x40x2 | 2 | ED1 | 0.82 | 0.01 | 181 | 0.02 | OK | D2, H1-1b |

# SERVICEABILITY AND STABILITY

*Table 9 - Serviceability and stability*

| Check | Value | Limit | Result |
|---|---|---|---|
| Shelf plate deflection (SLS) | 2.8 mm | L/125 = 4.9 mm (SBC 301 App. C) | **OK** |
| Front truss vertical deflection (SLS) | 2.8 mm | L/360 = 11.7 mm | **OK** |
| Top horizontal sway (SLS) | 1.0 mm | H/400 = 7.3 mm | **OK** |
| Uplift at any foot, 0.9D + 1.6N | none | no uplift | **OK** |

## CONCLUSIONS

**1.** The East Steel Frame ST-E2 with the East Arch Hangers, as built on 12 Nov 2026, satisfies SBC 306 strength (highest ratio 0.45) and SBC 301 serviceability.

**2.** The frame is stable standing on the floor without anchors or wall fixing under the SBC 301 / SBC 306 load combinations.

**3.** The hanger flat bars 40 x 6 are screwed to both arch heads at 150 mm centres; the M10 rods carry a nut and washer under and over the angle and are locked after levelling.

**4.** Robot model: Madina2_ST-E2_asbuilt.rtd. The Robot output note is issued separately as MD2-SF-ZZ-GF-CA-S-0002.